PINNGraPE
Physics-Informed Neural Network for Gravitational-waves Parameter Estimation

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Introduction

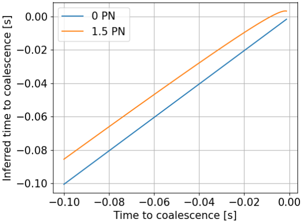
- **PINNGraPE** is a PyTorch algorithm which **does PE for a Gravitational-Wave (GW) signal's source thanks to a Physics-Informed Neural Network (PINN) [1].**
- **Motivations:**
  - LVK pipeline related:
    - cWB (one of the burst pipelines in LVK) need a PE component.
  - Theoretically related:
    - PINNs can approach (solving or doing PE) every system described by a differential equation (ex. TOV).

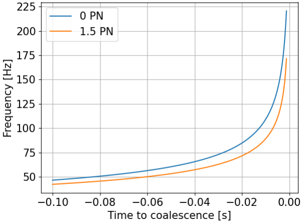
Physics-Informed Neural Networks

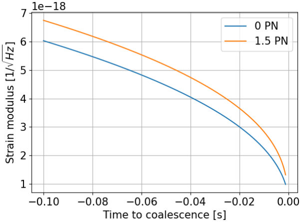
- Machine Learning techniques are algorithms built for engineering purposes, then applied to physics as they are.
- **PINNs** (Raissi et al. [1]) are NNs born in fluid dynamics to **solve or discover differential equations** behind data.

Training dataset

- **Data:** $\{(t_k, f_k), (t_k, t'_k), (t_k, |\tilde{h}_k|)\}$ points, where $t'_k = t[f(t_k)]$ and $\tilde{h}_k = \tilde{h}[f(t_k)]$.

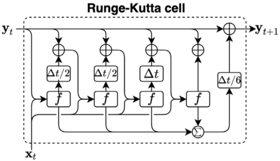






Algorithm basic structure: RNN

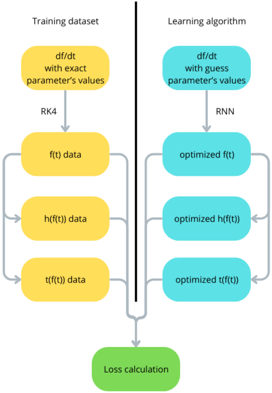
- We can impose (1), where $\mathcal{F}$ is a known functional and η and M_{tot} are the parameters to be inferred.
- **We solve (1) thanks to a Recurrent Neural Network** (RNN) with a Runge-Kutta integrator at 4$^{\text{th}}$ order implemented inside.



$$\frac{df}{dt} = \mathcal{F}[f, \eta, M_{tot}] \tag{1}$$

$$\mathcal{L} = \mathcal{L}\left[\sum_{k=\mathbf{1}}^{N}(f_k - f(t_k, \eta, M_{tot}))^{\beta}\right] + \dots \tag{2}$$

Structure and 1.5PN formalism



[2] + additional dimensional analysis

$$M_{tot} = m_1 + m_2, \quad \eta = \frac{m_1 m_2}{M_{tot}^2}, \tag{3}$$

$$\varepsilon = \frac{GM_\odot}{c^3}\left(\frac{M_{tot}}{M_\odot}\right)\pi f, \tag{4}$$

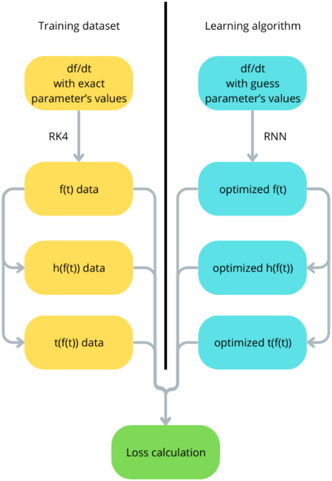
$$\frac{df}{dt} = \frac{96}{5}\pi^{8/3}\left(\frac{GM_\odot}{c^3}\right)^{5/3}\left(\frac{M_{tot}}{M_\odot}\right)^{5/3}\eta f^{11/3} \times \times \left[1 - \left(\frac{743}{336} + \frac{11}{4}\eta\right)\varepsilon^{2/3} + 4\pi\varepsilon\right] \tag{5}$$

$$t(f) = t_c - 5(8\pi f)^{-8/3}\left(\frac{GM_\odot}{c^3}\frac{M_{tot}}{M_\odot}\right)^{-5/3}\eta^{-1} \times \times \left[1 + \frac{4}{3}\left(\frac{743}{336} + \frac{11}{4}\eta\right)\varepsilon^{2/3} + \frac{32}{5}\pi\varepsilon\right] \tag{6}$$

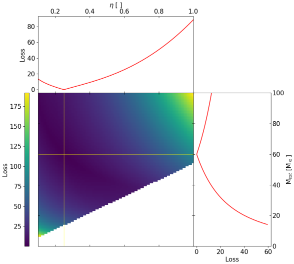
$$\Psi(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3}{4}\eta\left(8\pi\frac{GM_\odot}{c^3}\frac{M_{tot}}{M_\odot}f\right)^{-5/3} \times \times \left[1 + \frac{20}{9}\left(\frac{743}{336} + \frac{11}{4}\eta\right)\varepsilon^{2/3} - 16\pi\varepsilon\right] \tag{7}$$

$$\tilde{h}(f) = \frac{\chi c}{D}\sqrt{\frac{5}{24}}\pi^{-2/3}\left(\frac{GM_\odot}{c^3}\right)^{1/3}\left(\frac{M_{tot}}{M_\odot}\right)^{5/6}\eta^{1/2} f^{-7/6} e^{i\Psi(f)} \tag{8}$$

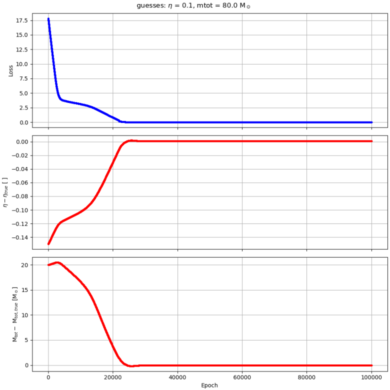
Structure and 1.5PN formalism

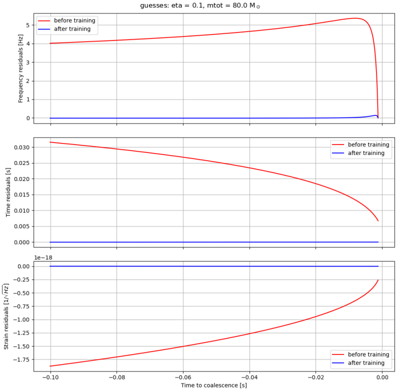


$$\mathcal{L} = \frac{\beta_f}{N}\sum_{k=1}^{N}|f_k - f(t_k)| + \\ + \frac{\beta_t}{N}\sum_{k=1}^{N}|t_k - t(f(t_k))| + \\ + \frac{\beta_h}{N}\sum_{k=1}^{N}|h_k - h(f(t_k))| \tag{9}$$



The best run (BBH)

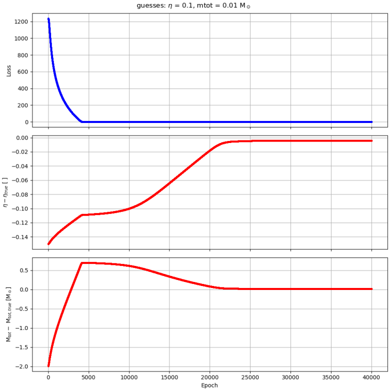


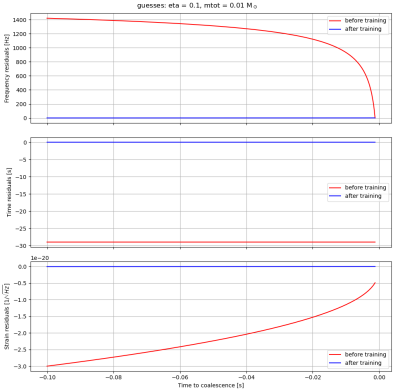


	η	M_{tot} $[M_\odot]$
Simulated data's values	0.2500	60.00
Guess values	0.1000	80.00
Inferred values	0.2507	59.95

- Final loss value: $2.385 \cdot 10^{-2}$.
- $\beta_f = 1$, $\beta_t = 0$, $\beta_h = 10^{19}$.
- AdamW optimizer, $lr_\eta = 10^{-4}$, $lr_{M_{tot}} = 10^{-2}$ (both fixed).
- Total time: $\sim$ 1 hour on CPU (4000 epochs).

The best run (BNS)





	η	M_{tot} $[M_\odot]$
Simulated data's values	0.2500	2.000
Guess values	0.1000	0.0100
Inferred values	0.2459	2.018

- Final loss value: $5.899 \cdot 10^{-2}$.
- $\beta_f = 1$, $\beta_t = 0$, $\beta_h = 10^{19}$.
- AdamW optimizer, $lr_\eta = 10^{-4}$, $lr_{M_{tot}} = 10^{-2}$ (both fixed).
- Total time: $\sim$ 1 hour on CPU (4000 epochs).

Conclusions and future prospects

- **PINNGraPE is able to infer η and M_{tot} values with 10^{-2} relative error** from frequency and strain data, implementing 1.5PN formalism.
- Still there is something that can be modified to have better performances (ex. adaptive learning rate).
- **Near future steps**:
  - to build a real dataset spanning a physical parameter space;
  - to test robustness against noise and glitches;
  - to extend the number of parameters to infer.
- (Not so) **remote future step**:
  - use of cWB real outputs,
  - apply PINNs approach to TOV equations, in order to constrain NS's equation of state.

Discussion

- Implementation of a training dataset spanning parameter space;
- PINNs formalism application on TOV equations;
- PINNs theoretical convergence for PE.

References I

[1] M. Raissi, P. Perdikaris, and G. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[2] C. Cutler and E. E. Flanagan, "Gravitational waves from merging compact binaries: How accurately can one extract the binary's parameters from the inspiral waveform?," *Physical Review D*, vol. 49, p. 2658–2697, Mar. 1994.

[3] R. G. Nascimento, K. Fricke, and F. A. Viana, "A tutorial on solving ordinary differential equations using python and hybrid physics-informed neural network," *Engineering Applications of Artificial Intelligence*, vol. 96, p. 103996, 2020.

[4] the LIGO Scientific Collaboration and the Virgo Collaboration, "The basic physics of the binary black hole merger gw150914," *Annalen der Physik*, vol. 529, Oct. 2016.

PINNGraPE_N

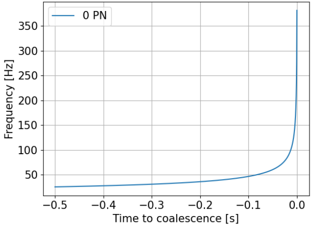
- **Data:** $\{(t_k, f_k), (t_k, h_k)\}$ points.
- **Optimizer:** Adam.
- **Learning rate:** Different for the two parameters.

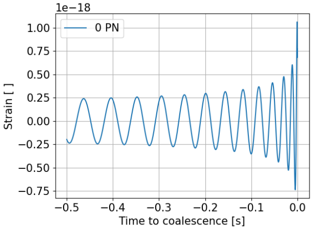
$$\frac{df}{dt} = \frac{96}{5}\pi^{8/3}\left(\frac{GM_\odot}{c^3}\right)^{5/3}\left(\frac{\mathcal{M}}{M_\odot}\right)^{5/3} 1^{\phi} f^{\alpha} \tag{10}$$

$$h(t) = \frac{4c}{D}\left(\frac{GM_\odot}{c^3}\right)^{5/3}\left(\frac{\mathcal{M}}{M_\odot}\right)^{5/3}(\pi f)^{2/3}\cos\left(2\pi f t + \phi\right)\cdot 10^{19} \tag{11}$$

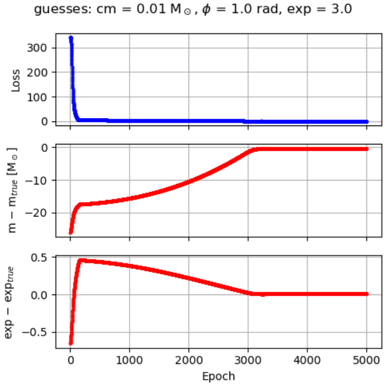
$$\mathcal{L} = \frac{1}{N}\sum_{k=1}^{N}|f_k - f(t_k)| + \frac{1}{N}\sum_{k=1}^{N}|h_k - h(t_k)| \tag{12}$$

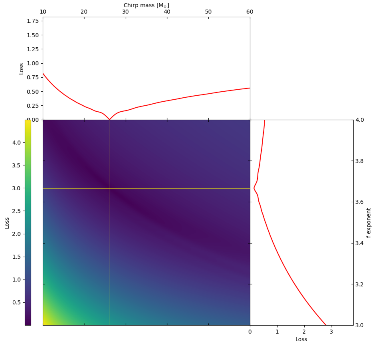
$$f_0 = \frac{1}{2\pi\sqrt{2}}\left(\frac{GM_\odot}{c^3}\right)^{-1}\left(\frac{M}{M_\odot}\right)^{-1} \quad \text{(fixed)} \tag{13}$$





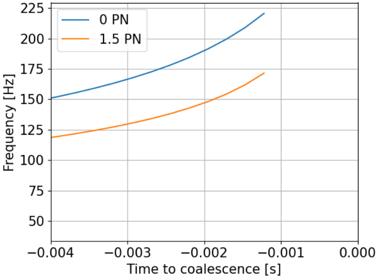
PINNGraPE_N


guesses: cm = 0.01 M$_\odot$, ϕ = 1.0 rad, exp = 3.0
Loss
m − m$_{true}$ [M$_\odot$]
exp − exp$_{true}$
Epoch


Chirp mass [M$_\odot$]
Loss
f exponent

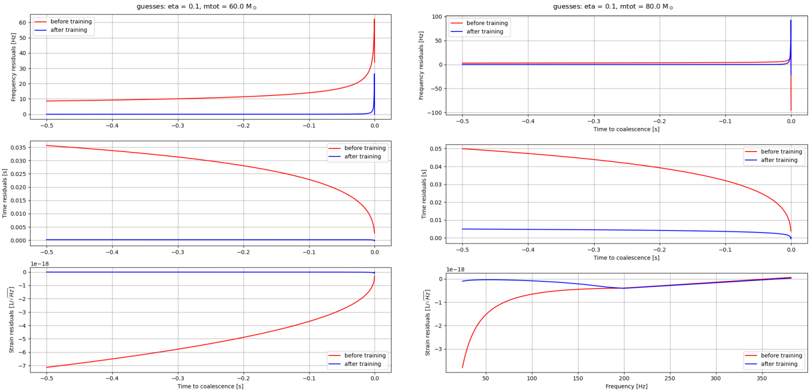
Main issues: hypothesis on systematic errors' cause

- Concerning the training dataset, **a starting RK4 with the Newtonian formalism must be implemented** to avoid divergences for the first integration point when working at 4kHz.
- **Solution:** create the data starting from f_0 with a Newtonian RK4 at 16kHz for $dt = 1/4096$ s, switching to a 1.5PN RK4 after this; then **cut the first few ms**.
- **Performance of the solution:** not systematically understood.

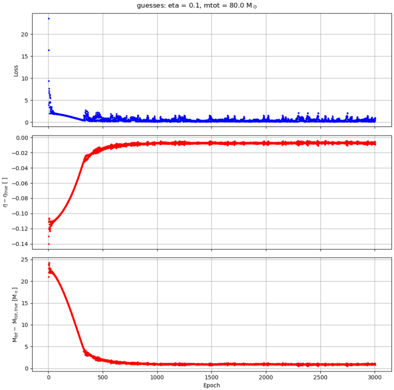


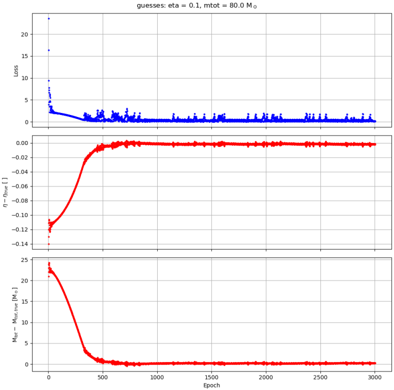
Main issues: hypothesis on systematic errors' cause

- Learning more then one parameter per time causes systematic errors on the inference.



Main issues: RK4 influence





Main issues: RK4 influence